

CTA Scenario Hypothetical - United Bank

Salesforce.com Certified Technical Architect Bootcamp

Mock Review Board Instructions

Candidate Prep Time: 90 minutes

Presentation Time: 30 minutes

Solution should cover:

- Actors and license choices
- System Landscape
- Role Hierarchy
- Data Model
- LDV
- App Design (Declarative and Programmatic Solutions)
- Integrations
- SSO
- Sharing Design
- Governance (if applicable)

Project Overview

United Bank (UB) has been in business for over 75 years selling financial products and services internationally. UB sells bank accounts, loans, insurance, credit cards and debit cards to individual customers as well as to businesses.

UB sells its financial service products through over 2,000 branches who are divided into 5 geographical locations worldwide. Each branch employs 10 staff on average and they service over 20M customers globally. There are currently 10M financial transactions per day across this customer base. UB expects to grow its footprint in Asia over the next 5 years and with this expansion expects the number of bank branches and customers to double.

Each of the 5 geographical locations within UB uses a proprietary CRM system with highly customised processes in place to work with customers that is expensive to change. UB would like to replace this system with a Salesforce solution. They also have a local 24x7 contact centre in each region to respond to customer enquiries by phone and email.

UB Branch Managers currently do not track a sales pipeline or a contact database of potential customers. UB Branch Staff have no means of seeing a complete picture of a customer's profile with the bank and are finding it hard to have meaningful conversations with their customer base.

Currently each UB Branch tracks needs analysis and applications manually using spreadsheets and the UB Branch Managers and the Executive team do not have good visibility into sales activity or customer interactions. UB would like to implement a Salesforce solution as a global system to manage all customer interactions, sales activity and provide a streamlined process to open new bank accounts.

UB Contact Centre Managers are looking to improve productivity and customer satisfaction by employing more effective ways of engaging with their customers. They have identified the need to be able to conduct web chats with customers, provide answers to Frequently Asked Questions and would like to crowd source answers to questions.

UB has identified 5 types of employees who will be using the system

- UB Executives oversee branch managers and would like to see sales revenue reports and sales pipeline reports by region
- UB Branch Staff are responsible for identifying sales deals and creating bank accounts
- UB Branch Managers are assigned a branch and are responsible for managing branch staff shift patterns and day to day running
- UB Contact Centre Agents are responsible for servicing customers needs via email and phone
- UB Contact Centre Manager oversee UB Banks contact centre and would like to see contact centre statistics reports

UB Banks customers will also need access to information stored in salesforce. Third party agents are also able to gather prospect data and are managed by UB Branch Managers.

The application must support English, Japanese, Chinese, French and German languages.

United Banks Systems

UB employees currently use multiple systems: a **core banking system**, an **enterprise service bus**, a **data warehouse** and an on premise **CRM system** and their enterprise architects have provided the following summary

Core Banking System

A mainframe system. The core banking system is state of the art and is the financial hub of the bank. It stores financial transactions, financial products and customer information. This system must be kept and exposes key functions via a SOAP API.

Enterprise Service Bus

A recent purchase which has seen limited use up until now. The CTO has mandated that it be used and must be kept. It supports REST, SOAP and XML interfaces and the bank is open to ideas on how best to leverage the technology.

Data Warehouse

An Oracle database used globally for reporting and aggregates data from numerous disparate sources. UBS employees access the system through the web using system-specific credentials.

CRM System

The CRM System is a .NET application and database that use SOAP API calls to retrieve and send information from/to the Core Banking System. UB would like to see the system replaced with salesforce and address the following problems:

- Prevent the duplication of customer and contact records
- There is no address validation leading to poor data quality
- Allow UB Branch Staff to more efficiently track prospects and potential deals
- UB Branch Staff currently have to login using a system specific username and password
- UB Branch Staff cannot create relationships between customers to see their family and work connections
- Creating a new bank account is a 15 step process and error prone
- No mobile options

UB would like to retire the CRM system and implement the following changes:

- Deduplicate and migrate all data into salesforce
- Create a single customer master for all customer records
- UB Branch Staff can create prospects and sales opportunities

- Provide real time address validation for newly created and updated customers
- Provide web chat functionality in the contact centres
- Provide FAQs in the contact centres for UB contact centre agents
- Provide an option to use mobile devices across the branch network to view their prospects, customers and opportunity data.
- Provide a recommendation for either a single or multiple Salesforce instances

As part of the architecture UB has mandated the use of single-sign on across all systems.

Sales Cycle

- UB Branch Staff schedule financial planning workshops through social media channels targeting a specific demographic of customers
- At the workshops, UB Branch Staff gather attendee information and create a prospect record with basic contact information
 - The system should ensure that customer or prospect information is not duplicated
 - In addition, the system should provide verify the customer's address meets postal standards
- UB Branch Staff call interested customers after the workshop and schedule an appointment for them to come into their nearest branch for a one-on-one needs analysis
 - Before the call the UB Branch Staff would like to view a map on their desktop/mobile device that displays the customers location and UB branches in the surrounding area
- UB Branch Staff then create a needs analysis that ask the customer a series of questions around 4 key sales areas: Insurance, Investment, Loan and Transaction indicating if they are long term or short term objectives. The needs analysis can be saved part way through and retrieved at a later date.
- The needs analysis should trigger the application to make financial product suggestions based on the customers answers to the key questions.
- The final step of the needs analysis is for the customer to indicate which recommendations they would like to progress. For any recommendations accepted it should trigger the application to automatically create sales opportunities
 - Following completion of the needs analysis the customer should receive a summary of the conversation to their email address as a pdf
 - Sales opportunities created should have an expected close date of 2 weeks from the current date and their sales status set to Open
- UB also uses third party agents to gather prospects and enter needs analysis records

Bank Account Application Process

- For deals that are for 'transaction' type financial products the UB Bank Staff can open an account on the spot.
- The UB Staff Member or the third party agent converts the prospect to a customer record
 - At this stage the prospect must supply identification documents passport and proof of address and these are recorded against the customer
 - If the third party agent converts the prospect, converted application should be assigned to to the UB Branch Manager that manages the region
- The UB Branch Staff create a bank account application capturing which customers (applicants) need access to the account and the bank account product identified during the needs analysis
 - The application status should be set to 'In Progress'
 - If any of the applicants are under 18 then the application should not be able to be submitted
 - A maximum of 5 applicants are allowed per bank account

- At any stage, the UB Staff Member can contact the applicant using a predefined email, which could include (i) requests for information, (ii) denial of application or (iii) other general matters.
- If the bank account application is for a foreign investor, it must be approved by the branch manager before it is allowed to progress further.
- The bank account application may be subject to review by government agencies, in which case the Core Banking System will cancel the application and update the account in the application. The UB staff Member is then notified by email.
- When the application details are complete the UB Branch Staff submits the account for processing
 - All application details should be sent to the Core Banking System
 - If the Core Banking System successfully creates the bank account then the application records status should be updated to 'Completed' and a Bank Account record should be created and associated to each applicant
 - If the Core Banking System returns an error then it should be displayed on the page immediately
 - Any application not submitted for 7 days should notify the UB Branch Staff member for follow up

Customer Servicing Process

- UB Contact Centre Agents need the ability to service customers via phone, email and web chat.
- UB Contact Centre Agents need the ability to search for customers using their UB Customer Number, 4 digits of Social Security Number and Bank Account Number
 - If the customer does not exist they need the ability to create a customer and send this information to the Core Banking System that will return a unique Customer Number
- Customers can raise service requests on their mobile device or portal:
 - This will be assigned to the appropriate regional contact center agent based on the type of service request raised and the capacity of the agent to handle that request
- UB Contact Centre Agents create service requests for inbound contact with customers and the UB Contact Centre Managers have defined Service Level Agreements on response times depending on the customer's priority level, which could be defined as Regular, Priority and VIP.
 - VIP customer's requests must be accepted within 1 day otherwise an email is sent to the regional manager. All other customers have a 2 day response window
- UB Contact Centre Agents need the ability to amend a customer address and have this validated in real time to ensure high data quality
 - If the customer has an 'Inactive' status then the Contact Centre Agent should be prevented from making changes to the record.
 - Updates to address details should be synchronized back to the Core Banking system
- UB Contact Centre Agents need the ability to search for bank account transactions that have been made to show credits and debits to their account
 - The system should provide for filtering by date range, account number and type
- UB Contact Centre Managers create Employee Discipline records that are confidential
- Customers have a requirement to access a portal to raise and manage their service requests, modify their profile data, view FAQs and crowd-source answers to questions.

Data Requirements

- UB would like to replicate all customer information from the Core Banking System to salesforce
- UB would like to replicate 15M+ bank accounts and their relationships to customers (30M+) from salesforce to the Core Banking System
- Customers, Bank Accounts and their relationships that are created directly in the Core Banking System should be updated into salesforce on a nightly basis

- Existing sales opportunities and prospects in spreadsheets should be made available in salesforce. Each branch currently has 2 spreadsheets with approximately 1000 opportunities and prospects. The spreadsheets contain large amounts of duplicate data and picklist values are not always standardized.
- Sales opportunities and prospects should be available in the data warehouse
- The system should ensure minimal duplication of customer information
- Approximately 10K needs analysis are created daily, with 20% of them progressing to opportunities
- UB would like to load 5MM historical service requests into Salesforce. Service requests grow at the rate of 5K per day.

Reporting Requirements

- UB Branch Staff and Third Party agents can view a report of needs analyses and applications
- UB Branch Executives would like a dashboard to track their top performing branch managers based on the volume and size of the applications closed
- Customers need to be able to view a history of their service requests they have raised
- UB Branch Managers need to be able to perform a trend analysis of how long applications spend in various sales stages over time

Accessibility Requirements

- Internal users are stored in Active Directory whereas Third Party Agents are stored in a home-grown LDAP store
- Internal users need to be automatically provisioned and deprovisioned
- Customers can self-register for access to the application

Security Requirements

- UB Branch Managers and UB Branch Staff should only see customers and related information in their geographic location
- Only UB Branch Managers and their executives can view agent information for the region they manage
- Only certain branch staff trained to do so can create applications for high risk products
- Third Party agents only have access to the needs analysis records that belong to them. Third party agents have a line manager at that agency that can also view these records.
- UB Branch Managers can view the needs analysis records for the third party agents they manage
- Contact Centre Agents can view all customers
- Contact Centre Agents can see all cases raised in their geographic location only
- Contact Centre Agents cannot delete customer records
- Employee discipline service requests must not be visible to Contact Centre Agents and only visible to Contact Centre Managers and UB Executives
- Social Security Number is considered confidential, should not be stored in clear-text and is only viewable by UB Branch Staff
- Customers need the ability to view FAQs and service requests they have raised
- Customers should not have the ability to see service requests raised by other customers
- UB Executives should see roll-up summary data across regions
- All web services need to be secured with 2-way SSL